A Hybrid Model of Vision Transformer and Skeleton-Based Features for Optimized Badminton Action Recognition

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Abstract

Badminton action recognition is crucial in sports analytics, enabling accurate classification of player movements for performance assessment and training optimization. This study addresses challenges in recognizing subtle variations in strokes and capturing the temporal dynamics of body movements. We propose a hybrid badminton action recognition system integrating Vision Transformer (ViTs) models with skeleton-based spatiotemporal feature analysis. The data set consists of video snippets, each containing a different badminton action. The preprocessing phase involves key frame extraction by selecting ten evenly spaced frames per video using linear interpolation. RGB features are obtained through a pre-trained ViTs model, which encodes global contextual information, while 3D positional landmarks are extracted using MediaPipe to capture player motion dynamics. These features are fused using concatenation and classified as a long-short-term memory (LSTM) network, which effectively models temporal dependencies in sequential data. Experimental results demonstrate that the fusion of RGB and skeletal data significantly improves recognition accuracy, achieving an accuracy of 98.68%, outperforming models trained on individual modalities. The findings highlight the advantage of hybrid models in leveraging the complementary strengths of different data representations for action recognition.

Keywords: Badminton action recognition, Vision Transformer, Skeleton-based analysis, Hybrid feature fusion, LSTM, Spatiotemporal analysis.

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1. Introduction

The field of action recognition in sports has greatly progressed by combining computer vision and machine learning techniques. However, badminton is a fast-paced and dynamic sport, but it is still quite difficult to recognize movement. Because badminton strokes are so complicated, accurate recognition and strong handling of complex motion patterns are essential [1, 2, 3]. In addition, previous research has shown ongoing problems with data reliability, as the recognition of badminton stroke technique is not always accurate [4]. Accurately classifying strokes is essential for player performance assessment and tactical analysis applications.

Badminton action recognition is a research field aimed at identifying and classifying movements in badminton. With the advancement of video processing technology and machine learning, action recognition has become increasingly accurate by utilizing spatio-temporal features, which combine spatial (position) and temporal (time) information. Skeleton data extraction techniques, such as MediaPipe and OpenPose, have opened new opportunities in action recognition. Skeleton data represent movement through 2D or 3D coordinates of body joint points, known as landmarks. Previous studies have used characteristics of the skeleton data to improve the recognition of badminton actions [1, 2, 3, 4]. By integrating spatio-temporal features from 3D skeleton data, accuracy improvements can be achieved, but there is still a lack of accuracy in the class of drive backhand [4].

In badminton action recognition, deep learning has been one of the core techniques for feature extraction tasks. Convolutional neural networks (CNN) are the leading models in this regard, which means that they gain full advantage of raw video frames without any manual intervention and can learn spatial and temporal patterns automatically. Liu et al. [3] present a novel deep-learning-based badminton action recognition model employing LSTM networks to make human skeleton sequences with multi-relational information. This study suggests that the LSTM-based approach mitigates execution costs and overgeneralization issues to some extent, achieving recognition accuracies of 63%, 84%, and 92% across different benchmark datasets. Rahmad et al. [5] developed a system to recognize badminton actions using pre-trained CNN models, including AlexNet, GoogleNet, VGG-16 and VGG-19. Their findings indicate that GoogleNet outperforms the other models, attaining the highest classification accuracy of 87.5%. These studies show that deep learning-based feature extraction from badminton action recognition is efficient, as previously stated. CNN can automatically learn action-related high-level features from video frames and therefore can be used for performance analysis and training use cases. However, these developments have some limitations, especially with regard to variability, visual occlusion in player movements, and lighting conditions that can severely degrade recognition performance.

Despite these advances, standalone ViTs and skeleton-based models exhibit inherent limitations. RGB-based models that use ViTs capture contextual and environmental cues but often struggle to isolate motion-specific information. In contrast, skeleton-based models accurately capture motion structure, but lack contextual understanding. This paper contributes to the recognition of badminton action through:

- A hybrid model that integrates vision transformer-based spatio-temporal RGB features with 3D skeleton-based spatio-temporal analysis.
- An improved classification accuracy.

2. Related Works

2.1. Skeleton-Based Action Recognition

Skeleton-based approaches for human action recognition have demonstrated remarkable performance in capturing spatial-temporal patterns, benefiting from advances in deep learning and graph-based methods. Traditional approaches initially relied on handcrafted features and statistical modeling, focusing on relative joint positions and angles to represent actions [6]. However, these methods struggled with modeling complex patterns, leading to the adoption of recurrent neural networks (RNNs) and LSTM networks, which improved temporal sequence modeling [7]. CNNs later gained popularity by encoding skeleton sequences into pseudoimages, enabling effective spatial feature extraction. The human actions are recognized using 3D skeleton data, data preprocessing, and a deep convolutional neural network (DCNN). It is compared to older DCNN models such as ResNet18 and MobileNetV2, which show better performance [8]. Further developments introduced Graph Convolutional Networks (GCNs) to capture relationships between joints, offering a more flexible and robust framework for processing non-Euclidean data [6, 9, 10, 11]. Adaptive approaches, such as the Adaptive Structure-Feature Fusion GCN (SFAGCN), have refined feature fusion strategies to balance spatial and temporal dependencies, improving classification performance [6]. Other advances, such as Action Capsules, focused on identifying key joints and modeling dynamic dependencies to improve action recognition under occlusion and missing data conditions [7]. For early action prediction, methods such as the adversarial-learning adaptive graph convolutional network (AGCN-AL) exploited partial sequences by aligning their representations with complete sequences using adversarial learning [9]. Furthermore, frameworks such as AlphaPose [12] and MediaPipe [13] provided practical implementations for pose estimation and feature extraction, supporting real-time performance in applications requiring large-scale human motion analysis. The introduction of RGB-D sensors also improved the recognition accuracy by using depth information and 3D skeleton data [14].

Skeleton-based approaches have been instrumental in advancing badminton action recognition by leveraging key point-based pose estimation techniques to model dynamic and complex motions. Liang et al. employed AlphaPose to extract skeleton data from badminton videos, demonstrating the effectiveness of LSTM networks in classifying actions with 80% accuracy, surpassing the 60% achieved by CNNs due to LSTM's ability to capture temporal dependencies [1]. Meanwhile, Asriani et al. performed spatio-temporal data extraction from 3D joint skeletons, utilizing Fast Dynamic Time Warping (FDTW) for temporal features and achieving 95.38% accuracy through classification with a weighted ensemble model combining SVM, LR, RF, and AdaBoost [4]. Erfianto et al. Improved performance using LSTM with landmark tracking to analyze badminton poses based on time series data, achieving over 90% precision for classifying smashes, serves and lobs [15]. These studies underscore the importance of skeleton-based data in representing spatial and temporal dynamics, allowing accurate recognition and classification of badminton actions for performance analysis and training systems.

2.2. Vision Transformer (ViTs)

ViTs have been widely applied in human action recognition, offering advantages over traditional CNN-based models. Unlike CNNs, which rely on local receptive fields, ViTs process image patches as sequences, allowing a more comprehensive understanding of global spatial dependencies [16, 17, 18]. Recent studies have successfully integrated ViTs with deep learning architectures to improve activity recognition, particularly in complex environments where motion occlusion and background clutter pose challenges. Han et al. [19] proposed an innovative ViT-based method integrated with AGCL to process 3D skeleton data, effectively combining spatial and temporal features for improved activity recognition. Similarly, Lamghari et al. introduced ActiViT, a ViT-based approach that dynamically selects key visual features, enabling it to focus on discriminative action patterns and outperform existing methods in handling complex actions [20]. Furthermore, Liu et al. integrated CNNs and ViTs, demonstrating that CNNs extract localized features. ViTs provide global contextual modeling, increasing the classification accuracy for gesture and activity recognition [21]. These advancements highlight the robustness of ViTs in processing multimodal inputs, including RGB and depth data, achieving state-of-the-art performance in benchmark data sets [17, 22, 23].

ViTs require pre-trained weights to achieve competitive performance, limiting their adaptability to domain-specific tasks [20, 22]. These limitations suggest the need for hybrid models that combine the strengths of CNNs and ViTs or lightweight variants that reduce computational costs while maintaining performance [21, 23].

2.3. Feature Fusion Techniques

By combining different data types to improve recognition performance, feature fusion techniques are crucial to the progress of action recognition systems. Much research has been done on combining spatial, temporal, and skeletal features in badminton action recognition to better understand how players move. Combining RGB-based features with 3D skeleton data has proven particularly effective due to their complementary nature. RGB features capture rich visual details such as texture and color, whereas 3D skeleton data focus on human motion's structural and kinematic properties. Putting these different types of information together improves the ability to recognize actions considering the context and space-time aspects [24]. For example, the Dense-Sparse Complementary Network (DSCNet) proposed by Cheng et al. [25] effectively leverages the complementary information of RGB and skeleton modalities through optimized sampling and a motion extraction module, resulting in better recognition accuracy at a reduced computational cost. The effectiveness of feature fusion lies in its ability to mitigate the individual limitations of different data modalities. Although efficient and robust against background noise, skeleton-based models often lack contextual information to distinguish similar actions. In contrast, RGB-based models struggle with environmental variations such as lighting changes and occlusions. Researchers have found ways to balance these pros and cons using attention mechanisms and multidimensional feature fusion networks [2, 26]. In particular, integrating spatiotemporal features using techniques such as Transformer-based networks and graph convolutional networks (GCNs) has significantly advanced state-of-the-art action recognition [27, 28]. However, challenges remain, particularly in optimizing the fusion process to handle variations in player movements and maintain high accuracy across diverse scenarios.

2.4. Hybrid Model in Action Recognition

Object detection plays a crucial role in human action recognition (HAR) by ensuring accurate identification of individuals before analyzing their movements. Deep learning-based methods such as YOLO have significantly improved human detection in intelligent surveillance systems (ISS). However, static confidence thresholds often fail to detect distant or occluded individuals, affecting the HAR's accuracy. Dynamic thresholding based on image geometry has been introduced to address this, enhancing detection reliability while reducing false positives [29]. Similarly, body part-based detection has improved feature extraction in surveillance applications [30] With enhanced human detection that ensures precise localization, advanced HAR models can leverage high-quality input features to enhance recognition performance.

Hybrid deep learning models are now very useful for recognizing human actions. They use the best parts of several deep learning architectures to improve accuracy and speed up computation. A common method combines CNN with RNN, such as Gated Recurrent Units (GRUs) and LSTM, to show how action sequences depend on each other both in space and time [31]. One such model, the Adaptive Hybrid Deep Attentive Network (AHDAN), combines a 1D-CNN with a GRU to extract meaningful features from multimodal sensor data and is further optimized using the enhanced archerfish hunting optimizer (EAHO), which improves classification accuracy while reducing computational complexity [32]. Another hybrid approach employs a duo-deep learning framework, where pre-trained MobileNet-V2 and DarkNet53 models extract features that are fused using a canonical correlation method, followed by an improved particle swarm optimization (PSO) algorithm to select the most relevant features for classification [33]. Additionally, incorporating multimodal data, such as skeleton-based features combined with RGB and depth information, has proven to enhance recognition robustness, particularly in real-world environments with occlusions and varying lighting conditions. These hybrid models perform much better than traditional single-network architectures. They show better recognition accuracy on a number of benchmark datasets, such as the KTH, UCF101, and Hollywood data sets [32]. As a result, hybrid deep learning models look like a good choice for fields such as healthcare, surveillance, and human-computer interaction that need to process information quickly and accurately.

2.5. Badminton Action Recognition

Methods for recognizing badminton actions can be categorized into two main approaches: machine learning-based and deep learning-based classifications. Ting et al. [34] employed SVM to classify badminton strokes into ten distinct types, demonstrating its effectiveness in badminton action recognition. Subsequent studies further validated the applicability of SVM in this domain. Anik et al. [35] investigated badminton stroke techniques, achieving an SVM classification accuracy of 88%. Similarly, Ghazali et al. [36] compared SVM with decision trees and KNN, reporting a superior precision of 83.4% for SVM. The study suggested that future research should focus on improving recognition accuracy through improved feature extraction techniques. Additionally, AdaBoost has been used for badminton action recognition, demonstrating higher recognition rates compared toHMM [37].

Deep learning has been applied in badminton action recognition. Wang et al. [38] described the use of the Adaptive Feature Extraction Block (AFEB)-AlexNet [38]. Rahmad et al. [39] compared four deep CNN models, namely AlexNet, GoogleNet, VGGNet-16, and VGGNet-19, with the best classification results achieved by GoogleNet. The classification results are high, but this study is weak because there are only two badminton stroke classes. Next, Rahmad et al. (2020) developed the number of stroke classes to five, including smash, clear, drop, lift, and net shot. With AlexNet, GoogleNet, VGGNet-16, and VGGNet-19 architectures, CNN was applied for feature extraction and SVM for classification. The results showed that GoogleNet provided the best accuracy in recognizing smash and non-smash techniques [29]. CNN and LSTM models are the primary architectures for analyzing video and skeletal data. CNN extracts spatial features from video frames, but LSTM can capture the temporal dynamics of sequential movements [1]. Graph Convolutional Networks (GCN) have been used to comprehend spatial-temporal correlations in skeletal data [3].

This research contributes to enhancing the accuracy of badminton action recognition by integrating RGB feature extraction using pre-trained ViTs models with 3D skeleton-based spatio-temporal analysis. Previous studies have demonstrated the effectiveness of deep learning in action recognition; however, they often struggle with capturing long-range dependencies in video sequences.

The introduction of ViTs addresses this limitation by leveraging self-attention mechanisms to extract global contextual information from RGB frames, thereby improving feature representation. Meanwhile, 3D skeleton-based spatio-temporal modeling has been widely used to capture human motion dynamics, offering robustness against variations in background and lighting conditions. By combining these approaches, this study aims to overcome the shortcomings of single-modal models and improve the recognition accuracy for complex and fast-paced badminton movements. This hybrid approach is expected to provide a more comprehensive representation of player actions, contributing to the advancement of automated sports analytics and intelligent coaching systems.

3. Method

Previous studies highlight persistent challenges related to data dependency, as evidenced by inconsistent accuracy in recognizing badminton stroke techniques. This study proposes a hybrid model that integrates ViTs with 3D skeleton position markers for feature extraction. A frame selection process is applied to video sequences, extracting only relevant frames. The selected frames are processed separately using ViT for spatial feature extraction and MediaPipe for skeletal feature extraction. The resulting features are then concatenated and classified using the LSTM network.

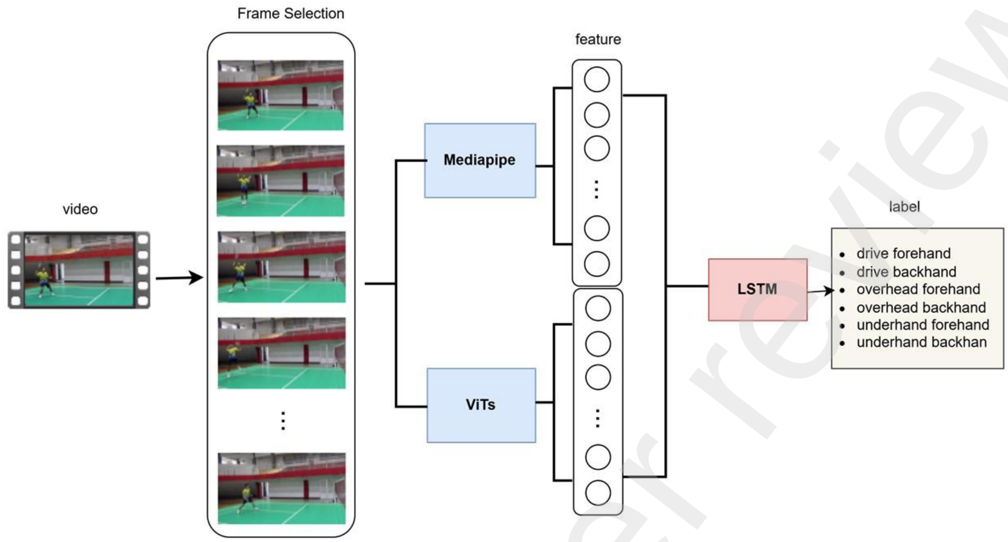

Figure 1: Proposed method

3.1. Data Acquisition

The dataset consists of video recordings captured at 30 frames per second using a 15MP DSLR camera in an indoor badminton court under controlled lighting conditions. Each video segment represents a different badminton stroke performed by PBSI-registered athletes between 15 and 22 years of age. 1,143 videos were used for training, while 380 videos were assigned for validation.

3.2. Frame Selection

Frame selection is crucial in video data processing, as it dictates which frames are chosen for subsequent analysis. Due to the dynamic nature of badminton movements, capturing the most representative frames embodying each action's spirit is crucial. The method used to select key frames is Linear Interpolation. The frames were chosen by dividing the range from the first to the last frame in equal parts, calculated using the formula in (1).

$$f_i = start + i \times \left(\frac{stop - start}{num - 1}\right) \tag{1}$$

The start is the first frame (0), the stop is the last frame (total frames - 1), num is the number of frames to be selected, and i is the index from 0 to num - 1. In this experiment, num=10 was chosen.

3.3. Feature Extraction

3.3.1. Mediapipe for feature extraction

MediaPipe is an open source framework developed by Google that enables the creation of multimodal real-time machine learning (ML) pipelines [10]. It supports the development of perceptual computing applications, including video analysis, gesture recognition, and facial detection. The MediaPipe algorithm estimates the positions of 33 body landmarks from RGB video frames. 3D pose extraction from 10 frames of badminton action images yields 990 features. Figure 2 illustrates the extraction of 3D skeleton poses using Mediapipe.

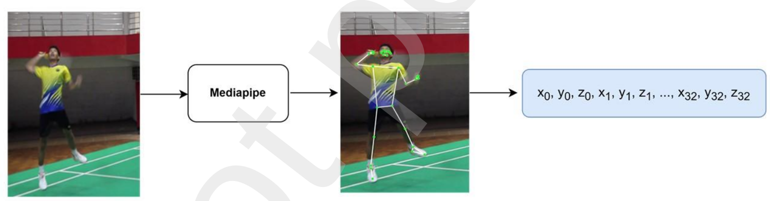

Figure 2: 3D skeleton feature extraction

3.3.2. Vision transformer for feature extraction (ViTs)

Feature extraction was performed using the ViTs model, specifically the pre-trained "google/vit-base-patch16-224-in21k" architecture. The input data consisted of video samples with varying durations. Each video was uniformly sampled to extract a fixed number of frames, which were resized to 224x224 pixels to match the input requirements of the ViT model. Figure 3 shows the ViTs model extracted feature vectors from each frame. Frame extraction using ViTs generates 768 features per frame, resulting in 7,680 features for 10 frames.

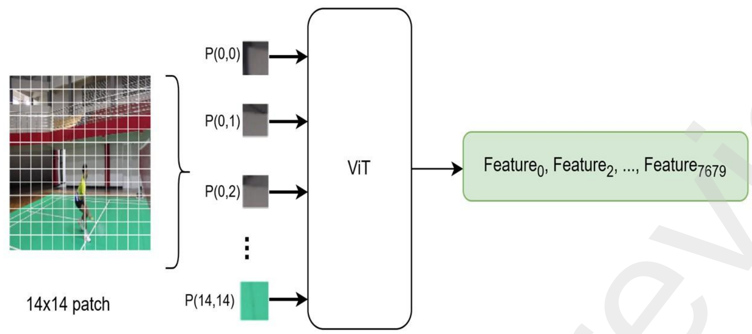

Figure 3: ViTs Feature extraction per frame

3.4. Feature Fusion

The feature concatenation process merges two distinct feature sets into a single vector. The first feature set (blue box) consists of 3D skeletal coordinates (x, y, z) from multiple key points, resulting in 990 features. The second feature set (green box) comprises 7,680 features extracted using a ViTs. The concatenation operation combines these feature sets into a single vector (gray box) with 8,010 features. Figure 4 illustrates the feature fusion with concatenation.

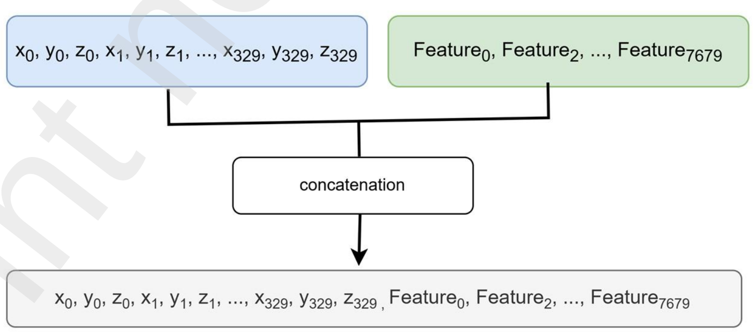

Figure 4: Concatenation feature fusion

3.5. LSTM for Sequence Classification

The LSTM architecture is presented in Figure 5. LSTM consists of an LSTM layer with n units using Tanh activation, a dense layer with m units using ReLU activation, and an output layer with SoftMax activation. Variations in the number of units n and m are detailed in the details of the experiment.

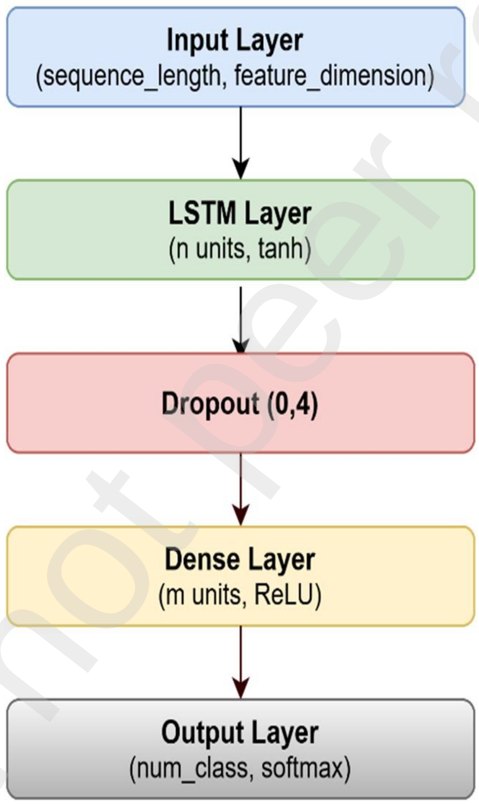

Figure 5: LSTM Architecture

4. Result and Discussion

We trained, tested, and evaluated the proposed method using three feature data sets: the 3D skeleton, the ViTs, and the integrated 3D skeleton and the ViTs data set. For each feature dataset, we compared our methodology to state-of-the-art techniques.

4.1. Detail Experiment

The experiment's results using 3D Skeleton data features are presented in Table 1, which shows the performance of various models in classifying badminton actions. Model3_A shows the best performance compared to other models, with the highest accuracy of 0.9526, precision of 0.9574, recall of 0.9593, and F1-score of 0.9532. With a 3D skeleton input feature dimension of 990, the optimal number of units is 275 LSTM units and 90 Dense units. This optimal configuration allows the model to better capture temporal patterns. The dropout rate of 0.3 is lower than the 0.4 used in most other models. A higher dropout rate may lead to the loss of important information, while a lower dropout rate helps the model retain more temporal information from the skeleton data.

Table 1: Evaluation of model with 3D Skeleton features

Model	Layers	Units per Layer	Dropout	Accuracy	Precision	Recall	F1-score
Model1_A	1 LSTM, 2 Dense	256 LSTM, 112 Dense	0.4	0.95	0.9517	0.95	0.9506
Model2_A	1 LSTM, 2 Dense	230 LSTM, 112 Dense	0.4	0.9342	0.9379	0.9342	0.9351
Model3_A	**1 LSTM, 2 Dense**	**275 LSTM, 90 Dense**	**0.3**	**0.9526**	**0.9574**	**0.9526**	**0.9532**
Model4_A	1 LSTM, 2 Dense	220 LSTM, 75 Dense	0.4	0.95	0.9542	0.95	0.9506
Model5_A	1 LSTM, 2 Dense	240 LSTM, 70 Dense	0.4	0.9289	0.9314	0.9289	0.9295
Model6_A	1 LSTM, 2 Dense	245 LSTM, 75 Dense	0.4	0.95	0.9526	0.95	0.9506

Table 2 displays the test results using ViTs-extracted data features, showing how well different models can classify badminton actions. Based on the test results,Model1_B performed better than other models, with the highest accuracy of 0.9684. In addition, this model has a precision of 0.9728, a recall of 0.9728, and an F1 score of 0.9688.Model1_B uses 256 LSTM units and 112 Dense units, larger than most other models. This combination of units allows the model to better capture spatio-temporal feature patterns from ViTs. In contrast, models with fewer units, such as Model5_B with 240 LSTM and 70 dense units, exhibit lower accuracy due to limited representation capacity.

Table 3 shows the test result that combined the data characteristics of ViTs and the Skeleton-Based Hybrid Model. Model5_AB achieved the highest accuracy of 0.9868 with a precision of 0.9874, a recall of 0.9868, and an F1 score of 0.9869. Model5_BA outperforms other models due to its optimal combination of LSTM and Dense units, a stable dropout rate of 0.4, and a well-balanced trade-off between complexity and generalization. This

Table 2: Evaluation of model with ViTs features

Model	Layers	Units per Layer	Dropout	Accuracy	Precision	Recall	F1-score
Model1_B	**1 LSTM, 2 Dense**	**256 LSTM, 112 Dense**	**0.4**	**0.9684**	**0.9728**	**0.9728**	**0.9688**
Model2_B	1 LSTM, 2 Dense	230 LSTM, 112 Dense	0.4	0.9184	0.9326	0.9184	0.9189
Model3_B	1 LSTM, 2 Dense	275 LSTM, 90 Dense	0.3	0.9289	0.9347	0.9289	0.9294
Model4_B	1 LSTM, 2 Dense	220 LSTM, 75 Dense	0.4	0.9553	0.9597	0.9553	0.9555
Model5_B	1 LSTM, 2 Dense	240 LSTM, 70 Dense	0.4	0.8921	0.915	0.8921	0.8909
Model6_B	1 LSTM, 2 Dense	245 LSTM, 75 Dense	0.4	0.9395	0.9421	0.9395	0.9396

model excels because 3D skeleton data directly capture body structure and movement patterns, while Vision Transformer (ViT) features provide visual information and motion context. Using ViTs and skeleton data effectively, Model5_BA achieves the highest classification accuracy compared to other models.

The experiments were conducted by swapping the order of the characteristics: Concatenate(Vision Transformer, 3D Skeleton). The highest accuracy obtained in this configuration was not as good as that of Model5_BA. The order of concatenation affects the way LSTM processes features. When 3D skeleton features are placed before RGB features, they become more dominant in capturing movement structure during LSTM processing. If the skeleton features are placed first (Model5_BA), the model can understand the structure of the movement and then enrich its understanding with the RGB features.

Table 3: Evaluation of model with hybrid ViTs and 3D skeleton features

Model	Layers	Units per Layer	Dropout	Accuracy	Precision	Recall	F1-score
Model1_AB	1 LSTM, 2 Dense	256 LSTM, 112 Dense	0.4	0.9474	0.9612	0.9474	0.9477
Model2_AB	1 LSTM, 2 Dense	230 LSTM, 112 Dense	0.4	0.9447	0.9576	0.9447	0.9447
Model3_AB	1 LSTM, 2 Dense	275 LSTM, 90 Dense	0.3	0.95	0.958	0.95	0.9503
Model4_AB	1 LSTM, 2 Dense	220 LSTM, 75 Dense	0.4	0.9763	0.9799	0.9763	0.9766
Model5_AB	**1 LSTM, 2 Dense**	**240 LSTM, 70 Dense**	**0.4**	**0.9868**	**0.9874**	**0.9868**	**0.9869**
Model6_AB	1 LSTM, 2 Dense	245 LSTM, 75 Dense	0.4	0.95	0.9615	0.95	0.9504

4.2. Experiment Analysis

The experimental results were analyzed by comparing the model with the highest accuracy for each data feature. Figure 6 illustrates the accuracy of each model. Model5_AB has the highest precision. Integrating 3D skeleton features with ViTs data improves precision. Figure 7 shows the confusion matrix for Model3_A, Model1_B, and Model5_AB. The amalgamation of features substantially improves accuracy. The drive backhand class had many classification mistakes, with 12 data points in Model3_A and 10 in Model1_B. Integrating data features significantly reduced misclassification errors, resulting in only two misclassified instances in the drive backhand category. In other classes, classification errors are managed effectively; however, in the Drive forehand class, the integration of features led to a rise in classification mistakes.

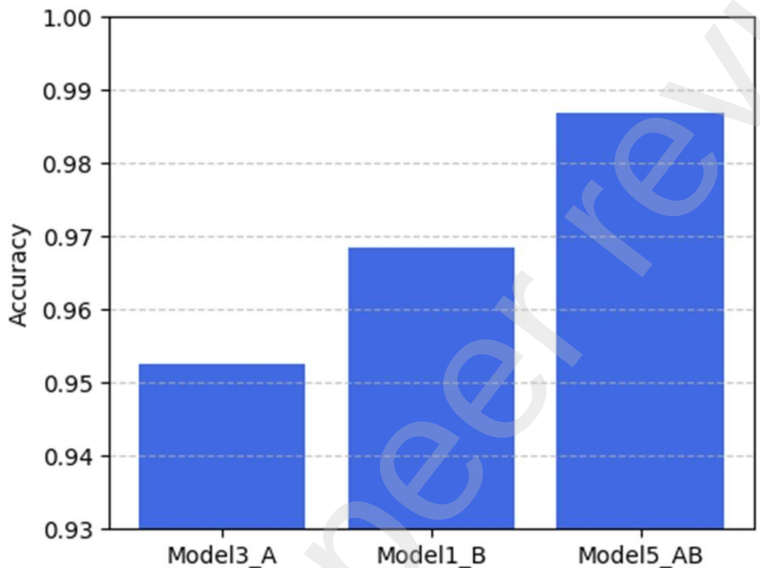

Figure 6: Accuration of model3_A, Model1_B, and Model5_AB

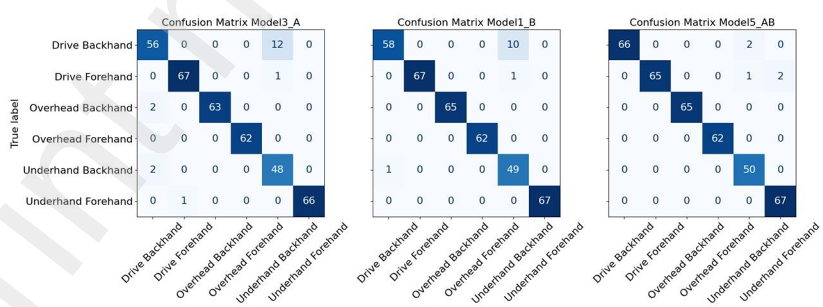

Figure 7: Confusion matrix of model3_A, Model1_B, and Model5_AB

Table 4 was developed from the comparison table in the previous study by Asriani et al. [4] by adding the top row that presents the Mediapipe-ViTs-LSTM method. This table compares the accuracy results between the proposed method and several previous studies in badminton action recognition, mainly using skeleton data and other sensor-based inputs. The proposed method, Mediapipe-ViTs-LSTM, achieved the highest performance with an accuracy of 98.68%, outperforming all other methods listed in the table. The extraction of spatiotemporal 3D skeleton supports the success of this method features using MediaPipe, combined with the powerful spatial representation capabilities of ViTs and the ability of LSTM to model temporal dynamics effectively.

Table 4: Comparation of results and state of the art

Method	Preprocessing Data	Akurasi (%)
Mediapipe-ViTs-LSTM	Spatiotemporal 3D skeleton by Mediapipe Spatiotemporal by ViTs	**98.68**
E2 (weighted ensemble SVM-LR-AdaBoost) [4]	Spatiotemporal 3D skeleton	95.38
LSTM [1]	Skeleton data extracted by AlphaPose	80.00
CNN [1]	Skeleton data extracted by AlphaPose	60.00
PDDRNet - XGBOOST [2]	Human pose joint skeleton	93.50
GCN [18]	Human Skeleton Sequence	92.00
SVM [34]	RGB-D sensors	92.00
SVM [35]	Accelerometer and gyroscope	88.89
SVM [36]	Inertial sensor	83.40
HMM [40]	Bounding box, HOG	83.33

5. Conclusion

This study presents a hybrid badminton action recognition model that combines ViTs-based RGB feature extraction with 3D skeleton-based spatiotemporal analysis. Integrating these features using an LSTM classifier significantly improves recognition accuracy, achieving a performance of 98. 68%, outperforming previous approaches. The findings highlight the potential of multimodal fusion in sports action recognition, providing valuable insights for automated coaching systems and performance analytics.

Future work will focus on optimizing computational efficiency and extending the approach to real-time applications using edge AI hardware. Further investigations of attention-based feature fusion techniques may improve interpretability and robustness.

6. Acknowledgment

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References

[1] Z. Liang and T. E. Nyamasvisva, "Badminton action classification based on human skeleton data extracted by alphapose," in *Int. Conf. on Sens. Meas. Data Anal. in the Era of Artif. Intell.*, 2023.

[2] X.-W. Zhou, L. Ruan, S.-S. Yu, J. Lai, Z.-F. LI, and W.-T. Chen, "Badminton action classification based on pddrnet," in *3rd Int. Conf. on Internet, Educ. Inf. Technol.*, 2023, pp. 980–987.

[3] J. Liu and B. Liang, "An action recognition technology for badminton players using deep learning," *Mobile Inf. Syst.*, vol. 2022, pp. 1–10, 2022.

[4] F. Asriani, A. Azhari, and W. Wahyono, "Improving badminton action recognition using spatio-temporal analysis and a weighted ensemble learning model," *Computers, Materials and Continua*, vol. 81, pp. 3079–3096, 2024.

[5] N. A. Rahmad, N. A. J. Sufri, M. A. As'Ari, and A. Azaman, "Recognition of badminton action using convolutional neural network," *Indonesian J. of Elect. Eng. Inf.*, vol. 7, pp. 750–756, 12 2019.

[6] Z. Zhang, Z. Wang, S. Zhuang, and F. Huang, "Structure-feature fusion adaptive graph convolutional networks for skeleton-based action recognition," *IEEE Access*, vol. 8, pp. 228 108–228 117, 2020.

[7] M. Liu, C. Chen, F. Meng, and L. Hong, "3d action recognition using multi-temporal skeleton visualization," in *International Conference on Multimedia and Expo (ICME.* IEEE, 2017.

[8] N. Tasnim, M. Islam, and J. H. Baek, "Deep learning-based action recognition using 3d skeleton joints information," *Inventions*, vol. 5, pp. 1–15, 9 2020.

[9] A. F. Bavil, H. Damirchi, and H. D. Taghirad, “Action capsules: Human skeleton action recognition,” *Computer Vision and Image Understanding*, vol. 233, 8 2023.

[10] H. Pei, J. Chen, S. Gao, T. Jin, and K. Lu, “Skeleton action recognition via group sparsity constrained variant graph auto-encoder,” *Image and Vision Computing*, vol. 154, 2 2025.

[11] Y. Xing and J. Zhu, “Deep learning-based action recognition with 3d skeleton: A survey,” *CAAI Transactions on Intelligence Technology*, vol. 6, pp. 80–92, 3 2021.

[12] H. S. Fang, J. Li, H. Tang, C. Xu, H. Zhu, Y. Xiu, Y. L. Li, and C. Lu, “Alphapose: Whole-body regional multi-person pose estimation and tracking in real-time,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, pp. 7157–7173, 6 2023.

[13] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann, “Mediapipe: A framework for building perception pipelines,” 6.

[14] B. Liu, H. Cai, Z. Ju, and H. Liu, “Rgb-d sensing based human action and interaction analysis: A survey,” *Pattern Recognition*, vol. 94, pp. 1–12, 10 2019.

[15] B. Erfianto, B. Purnama, and I. R. Wirawan, “Time series classification of badminton pose using long short-term memory with landmark tracking,” *Journal of Electronics, Electromedical Engineering, and Medical Informatics*, vol. 7, pp. 27–37, 1 2025.

[16] C. Wu and T. He, “A survey of applications of vision transformer and its variants,” in *Proceedings - 2024 IEEE 10th International Conference on Intelligent Data and Security, IDS 2024*. Institute of Electrical and Electronics Engineers Inc., 2024, pp. 21–25.

[17] R. Kumar and S. Kumar, “Effectiveness of vision transformers in human activity recognition from videos,” in *2023 International Conference on Advancement in Computation and Computer Technologies, InCACCT 2023*. Institute of Electrical and Electronics Engineers Inc., 2023, pp. 593–597.

[18] A. Berroukham, K. Housni, and M. Lahraichi, "Vision transformers: A review of architecture, applications, and future directions," in *Colloquium in Information Science and Technology, CIST*. Institute of Electrical and Electronics Engineers Inc., 2023, pp. 205–210.

[19] H. Han, H. Zeng, L. Kuang, X. Han, and H. Xue, "A human activity recognition method based on vision transformer," *Scientific Reports*, vol. 14, 12 2024.

[20] S. Lamghari, G. A. Bilodeau, and N. Saunier, "Transformer-based human action recognition with dynamic feature selection," in *Proceedings - 2023 20th Conference on Robots and Vision, CRV 2023*. Institute of Electrical and Electronics Engineers Inc., 2023, pp. 129–136.

[21] X. Liu, L. Hu, L. Tie, L. Jun, X. Wang, and X. Liu, "Integration of convolutional neural network and vision transformer for gesture recognition using semg," *Biomedical Signal Processing and Control*, vol. 98, 12 2024.

[22] F. Luo, S. Khan, B. Jiang, and K. Wu, "Vision transformers for human activity recognition using wifi channel state information," *IEEE Internet of Things Journal*, vol. 11, pp. 28 111–28 122, 2024.

[23] J. Wensel, H. Ullah, and A. Munir, "Vit-ret: Vision and recurrent transformer neural networks for human activity recognition in videos," *IEEE Access*, vol. 11, pp. 72 227–72 249, 2023.

[24] R. Yue, Z. Tian, and S. Du, "Action recognition based on rgb and skeleton data sets: A survey," *Neurocomputing*, vol. 512, pp. 287–306, 11 2022.

[25] Q. Cheng, J. Cheng, Z. Liu, Z. Ren, and J. Liu, "A dense-sparse complementary network for human action recognition based on rgb and skeleton modalities," *Expert Systems with Applications*, vol. 244, 6 2024.

[26] M. A. khan, M. Sharif, T. Akram, M. Raza, T. Saba, and A. Rehman, "Hand-crafted and deep convolutional neural network features fusion and selection strategy: An application to intelligent human action recognition," *Applied Soft Computing Journal*, vol. 87, 2 2020.

[27] X. Xiang, X. Li, X. Liu, Y. Qiao, and A. E. Saddik, "A gcn and transformer complementary network for skeleton-based action recognition," *Computer Vision and Image Understanding*, vol. 249, 12 2024.

[28] H. Wu, X. Ma, and Y. Li, "Transformer-based multiview spatiotemporal feature interactive fusion for human action recognition in depth videos," *Signal Processing: Image Communication*, vol. 131, 2 2025.

[29] Wahyono, M. E. Wibowo, A. Ashari, M. P. K. Putra, G. Mada, and Y. Indonesia, "Improvement of deep learning-based human detection using dynamic thresholding for intelligent surveillance system," p. 2021. [Online]. Available: www.ijacsa.thesai.org

[30] Wahyono, J. Hariyono, and K. H. Jo, "Body part boosting model for carried baggage detection and classification," *Neurocomputing*, vol. 228, pp. 106–118, 3 2017.

[31] N. Jaouedi, N. Boujnah, and M. S. Bouhlel, "A new hybrid deep learning model for human action recognition," *Journal of King Saud University - Computer and Information Sciences*, vol. 32, pp. 447–453, 5 2020.

[32] A. Y. A. B. Ahmad, J. Alzubi, S. James, V. O. Nyangaresi, C. Kutralakani, and A. Krishnan, "Enhancing human action recognition with adaptive hybrid deep attentive networks and archerfish optimization," *Computers, Materials and Continua*, vol. 80, pp. 4791–4812, 2024.

[33] M. N. Akbar, F. Riaz, A. B. Awan, M. A. Khan, U. Tariq, and S. Rehman, "A hybrid duo-deep learning and best features based framework for action recognition," *Computers, Materials and Continua*, vol. 73, pp. 2555–2576, 2022.

[34] H. Y. Ting, K. S. Sim, and F. S. Abas, "Automatic badminton action recognition using rgb-d sensor," *Adv. Mater. Res.*, vol. 1042, pp. 89–93, 2014.

[35] M. A. I. Anik, M. Hassan, H. Mahmud, and M. K. Hasan, "Activity recognition of a badminton game through accelerometer and gyroscope," in *Int. Conf. on Comput. Inf. Technol.* IEEE, 2 2017, pp. 213–217.

[36] N. F. Ghazali, N. Shahar, and M. A. As'ari, "Badminton strokes recognition using inertial sensor and machine learning approach," in *Int. Conf. on Intell. Cybern. Technol. Appl.* IEEE, 2022, pp. 1–5.

[37] J. Long and S. Rong, "Application of machine learning to badminton action decomposition teaching," *Wireless Commun. Mob. Comput.*, vol. 2022, 2022.

[38] Y. Wang, W. Fang, J. Ma, X. Li, and A. Zhong, "Automatic badminton action recognition using cnn with adaptive feature extraction on sensor data," *ICIC*, vol. 11643 LNCS, pp. 131–143, 2019.

[39] N. A. Rahmad and M. A. As'ari, "The new convolutional neural network (cnn) local feature extractor for automated badminton action recognition on vision-based data," in *Int. Conf. on Emerg. Comput. Technol. Sport.*, vol. 1529, 6 2020.

[40] W. T. Chu and S. Situmeang, "Badminton video analysis based on spatiotemporal and stroke features," in *Proc. of Int. Conf. on Multimedia Retrieval*, 2017, pp. 448–451.